

# **AMAZON SALES DATA ANALYTICS**

## **Executive Analysis & Business Intelligence Report**

### **ApexPlanet Internship – Data Analytics Project**

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**Technology:** Python • SQL • Power BI •  
Statistics • Machine Learning

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# Executive Summary

This project presents an end-to-end analysis of an Amazon Sales dataset using Python, SQL, Power BI, statistical analysis, and machine-learning techniques. The objective was to transform raw transactional data into meaningful business insights that can support sales monitoring, operational analysis, and data-driven decision-making.

The project began with data cleaning and exploratory data analysis. Missing values, duplicate records, inconsistent data types, column names, and potential outliers were examined and addressed. Statistical summaries and visualizations were then used to understand order quantities, order amounts, product categories, order statuses, and geographical patterns.

SQL was used to perform business-oriented analysis including monthly sales trends, category performance, cumulative revenue, moving averages, and aggregated sales metrics. Python was integrated with the database to automate data extraction and analysis.

Interactive dashboards were developed using Power BI to present key performance indicators, monthly revenue trends, category performance, order distributions, geographical performance, filters, and drill-through analysis.

Statistical analysis included descriptive statistics, hypothesis testing, correlation analysis, stationarity testing, and predictive modeling. Regression and classification models were evaluated using appropriate performance metrics.

The analysis identified differences in revenue across product categories, variations in monthly sales, order-status patterns, and potential outliers in order amounts. However, the dataset contains only four months of monthly observations and does not provide a reliable unique customer identifier. Therefore, long-term seasonal forecasting and customer-level churn or retention analysis have important limitations.

Overall, the project demonstrates a complete data analytics workflow—from raw data preparation to SQL analysis, visualization, statistical modeling, and business recommendations.

# Problem Statement

Organizations generate large volumes of transactional sales data, but raw data alone does not provide actionable business insight. The objective of this project is to analyze Amazon sales data to identify sales trends, product performance, order patterns, geographical differences, and statistical relationships that can support business decision-making.

## Objectives

- Clean and preprocess the raw sales dataset.
- Perform exploratory data analysis.
- Extract business insights using SQL.
- Create interactive dashboards.
- Analyze statistical relationships and patterns.
- Apply predictive modeling techniques.
- Identify anomalies and important trends.
- Provide actionable business recommendations.
- Document limitations and opportunities for future analysis.

# Amazon Sales Dataset

| Attribute         | Description                 |
|-------------------|-----------------------------|
| Data type         | Transactional sales data    |
| Main target       | Order amount / order status |
| Time dimension    | Order date                  |
| Product dimension | Category                    |
| Quantity          | Number of items ordered     |
| Geography         | Shipping state              |
| Status            | Order fulfilment status     |

## Methodology

Raw Data



Data Cleaning



Exploratory Data Analysis



SQL Analysis



Python Visualization



Power BI Dashboard



Statistical Analysis



Predictive Modeling



Business Insights



Recommendations

## Data Cleaning & EDA

- Missing-value handling
- Duplicate removal
- Data-type conversion
- Date conversion
- Categorical conversion
- Column-name standardization
- Outlier detection using IQR
- Statistical summaries

### Example finding

Order quantities are concentrated around smaller values, while a smaller number of observations contain higher quantities. Order amount analysis also identified potential outliers that were investigated using the IQR method.

## SQL Analysis

### Key analyses

## Monthly sales

```
SELECT
    strftime('%Y-%m', date) AS month,
    SUM(amount) AS total_revenue
FROM amazon_sales
GROUP BY month
ORDER BY month;
```

## Category performance

```
SELECT
    category,
    COUNT(*) AS total_orders,
    SUM(qty) AS total_quantity,
    SUM(amount) AS total_revenue,
    AVG(amount) AS average_order_value
FROM amazon_sales
GROUP BY category
ORDER BY total_revenue DESC;
```

- Moving average
- Cumulative revenue
- Ranking
- Window functions
- EXPLAIN QUERY PLAN
- Indexing

## Dashboard Section

### KPI Cards

- Total Revenue

- Total Orders
- Total Quantity
- Average Order Value

## **Visuals**

- Monthly Revenue Trend
- Category Revenue
- Order Status Distribution
- State Performance
- Top Products
- Filter/Slicer panel

## **Statistical Analysis**

| <b>Analysis</b>    | <b>Purpose</b>               | <b>Result</b>         |
|--------------------|------------------------------|-----------------------|
| Mean               | Average order amount         | 648.5614647631621     |
| Median             | Central tendency             | 605.0                 |
| Standard deviation | Variability                  | 281.21168722916195    |
| Skewness           | Distribution shape           | 0.8854806405409467    |
| t-test             | Compare mean with benchmark  | 183.90277421166792    |
| Chi-square         | Test categorical association | 107980.66171896397    |
| ANOVA              | Compare categories           | 9715.167517877611     |
| ADF                | Test stationarity            | 7.580290515681566e-21 |

## Predictive Modeling

### Regression

Models:

- Linear Regression
- Decision Tree Regression

Metrics:

- $R^2$
- MAE
- RMSE

## **Classification**

Models:

- Logistic Regression
- Decision Tree Classifier

Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

# **Top 7 Business Insights**

## **1. Revenue varies across months**

Monthly analysis shows changes in revenue over the available period, highlighting periods of relatively higher and lower sales activity.

## **2. Product categories contribute differently to revenue**

Category-level analysis shows that some categories contribute substantially more revenue than others.

## **3. Order quantities are concentrated around smaller values**

Most transactions involve relatively small quantities, while a smaller number of orders contain unusually high quantities.

## **4. Order amounts contain potential outliers**

IQR-based analysis identified unusually large or small order amounts that require consideration during business analysis.

## **5. Order statuses show different fulfilment outcomes**

The dataset contains multiple fulfilment states, including shipped, delivered-to-buyer, cancelled, returned, and pending orders.

## **6. Geographical performance differs**

Revenue and order activity vary across shipping states, indicating geographical differences in sales performance.

## **7. Dataset limitations affect forecasting and customer analysis**

Only four months of monthly observations are available, limiting reliable seasonal forecasting. The absence of a unique customer identifier also prevents reliable customer-level churn and retention analysis.

# **Recommendations**

## **Recommendation 1 — Focus on high-performing categories**

Prioritize inventory availability and marketing for categories that consistently generate higher revenue.

## **Recommendation 2 — Monitor cancellation and return rates**

Track order-status patterns regularly to identify categories, locations, or periods with unusually high cancellations or returns.

### **Recommendation 3 — Improve inventory planning**

Use monthly sales trends and category-level demand patterns to support inventory allocation.

### **Recommendation 4 — Track order-value outliers**

Large transactions should be monitored separately because they can significantly influence overall revenue metrics.

### **Recommendation 5 — Collect customer identifiers**

A reliable customer ID would enable:

- Customer segmentation
- Retention analysis
- Churn prediction
- Customer lifetime value
- Repeat-purchase analysis

### **Recommendation 6 — Collect longer historical data**

At least **12–24 months** of data would significantly improve:

- Seasonal analysis
- Trend detection
- ARIMA/SARIMA forecasting

## **Future Scope**

Future development could incorporate customer-level identifiers, product-level historical information, promotional data, discounts, and longer time-series observations. Advanced forecasting models could then be developed using larger datasets. Customer segmentation and churn prediction could also be performed once reliable customer-level information becomes available.

The analytics pipeline could additionally be automated so that new sales data is automatically cleaned, loaded into the database, analyzed, and reflected in the dashboard.

# Limitations

1. The dataset contains only a limited historical period.
2. Only four monthly observations are available, making reliable annual seasonality analysis difficult.
3. Long-term ARIMA/SARIMA forecasting is therefore limited.
4. No reliable unique customer identifier is available.
5. Customer churn and retention cannot be directly calculated.
6. Customer-level segmentation cannot be reliably performed.
7. Some potentially important variables such as customer demographics, promotions, discounts, and marketing information are unavailable.
8. Predictive models should therefore be interpreted as exploratory rather than production-ready models.

# Final Conclusion

This project demonstrated an end-to-end data analytics workflow using Python, SQL, Power BI, statistics, and machine learning. The analysis transformed raw Amazon sales data into structured information and actionable business insights.

Through data cleaning, exploratory analysis, SQL querying, visualization, dashboard development, statistical testing, and predictive modeling, the project provided a comprehensive view of sales performance and order behavior.

The project also highlighted the importance of understanding data limitations. In particular, the short historical period and lack of a unique customer identifier restrict long-term forecasting and customer-level analysis.

With additional historical and customer-level data, the analytical pipeline could be extended into a more advanced business intelligence and predictive analytics solution.